

68-XT1404 - INSPECTION AND TESTS PLANS FOR TURBO EXPANDER TC6-500 FOR
UPGRADER AND EXPANDER / HELIUM LIQUEFIER / DISTRIBUTION SYSTEM

FINAL

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DATE: xxx.xx.xx	

50	FN (Final)	F. MENY	N. TRONQUOY	O. LAPIERRE	17/03/2023
0	AFC (Approved for construction)	R. FRANGE	N. TRONQUOY	O. LAPIERRE	25/01/2023
B	IFR (Issued for review)	R. FRANGE	S. CREVATIN	O. LAPIERRE	21/12/2022
REV	DESCRIPTION	BY	CHKD	APVD	DATE
REVISIONS					
CLIENT : Qatargas Operating Company Limited			CONTRACT No. : LTC/C/NFE/3945/18		
PROJECT : NORTH FIELD EXPANSION Project - HeRU 4 EP					
DOC CLASS : 2					
DOC NO. : 3945_18-ME20829A-00452					

Template: M20-F21 (0)

Operation Center Document No.: 53125-RE-050-313F733



●BUSINESS CONFIDENTIAL
INFORMATION

INSPECTION PLAN

AIR LIQUIDE ADVANCED
TECHNOLOGIES

Doc. No: **C1377-IP-106 (2)**

Cust. No: **53125-RE-050-313F733**

Project No: **PRJ000614 - C1377**

Page: **2 / 3**

QATAR 4 EP

68-XT1404 - INSPECTION AND TESTS PLANS FOR TURBO EXPANDER TC6-500 FOR UPGRADER AND EXPANDER / HELIUM LIQUEFIER / DISTRIBUTION SYSTEM

WARNING :

THIS DOCUMENT AND THE INFORMATION IT CONTAINS ARE CONFIDENTIAL AND BELONG TO AIR LIQUIDE ADVANCED TECHNOLOGIES (« AL-aT »). THEY SHALL NOT BE USED FOR ANY PURPOSE OTHER THAN THOSE FOR WHICH IT WAS DISCLOSED. THEY SHALL NOT BE REPRODUCED OR DISCLOSED (IN WHOLE OR IN PART) TO ANY THIRD PARTY WITHOUT AL-aT'S PRIOR WRITTEN CONSENT.

Document No. (Revision)	Date	Distribution
C1377-IP-106 (2)	2022-12-19T00:00:00Z	- Project team - Customer - Suppliers / Integrators
Author	Checking	Approval
François MENY	Nicolas TRONQUOY	Olivier LAPIERRE
Visa	Visa	Visa
Refer to E-Signature on cover page of ITP	Refer to E-Signature on cover page of ITP	Refer to E-Signature on cover page of ITP

Template: M20-F21 (0)



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INSPECTION PLAN

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Page: **3 / 3**

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REVISION HISTORY

This document supersedes the following document(s):

Revision	Date	Description of Change	Author
0	09/12/2022	First release	R. FRANGE
1	23/01/2023	according to PIM meeting	R. FRANGE
2	17/03/2023	numbering mistake on manufacturing number of equipped part	F. MENY



HeRU 4 : Helium Liquefier and Distribution System

R2-05-F38 (3)

Plan d'inspections et de Tests ITP - Inspection and Test Plan**HeRU 4 (NFE ONSHORE HELIUM
RECOVERY UNIT)
TC6-500 T4**

Client - Customer : Qatargas Operating Company Limited
(ALAT) N° du Projet - (ALAT) Job N° : PRJ000614
Racine Projet C4168
N° de fabrication- Manufacturing N° : Equipped part : ALAT 2301520 (criticality level IIc)
Main cartridge : TC6-639-HI1 (criticality level IIc)
Spare cartridge : TC6-642-HI1 (criticality level IIc)

Approuvé par / Approved by: <i>Project Quality Manager</i>	Diffusion : Customer
Nom : Olivier LAPIERRE Date : LAPIERRE- Visa : SC, Olivier Digitally signed by LAPIERRE-SC, Olivier Date: 2023.03.20 11:19:13 +01'00'	ALAT Internal Production coordinator
Vérifié par / Verified by: <i>Manufacturing Engineer</i>	
Nom : Nicolas TRONQUOY Date : TRONQUOY, Visa : Nicolas Digitally signed by TRONQUOY, Nicolas Date: 2023.03.20 10:06:54 +01'00'	
Rédigé par / Prepared by: <i>Project Manager</i>	
Nom : François MENY Date :  Visa :  Signature numérique de François Meny Date : 2023.03.20 09:09:07 +01'00'	

ENREGISTREMENT DES MODIFICATIONS

MODIFICATIONS RECORDS

RELEASE	DATE	PAGES OR CHAPTERS MODIFIED	MODIFICATIONS JUSTIFICATION
0	9/12/2022	NA	First released
1	23/01/2023	ALL	according to PIM meeting
2	17/03/2023	ALL	numbering mistake on manufacturing number of equipped part

Acronymes :

AL-AT : Air Liquide
FAB : AL-AT Manufacturing
AQ : Assurance Qualité Fabrication AL-AT
AL GECS : Air Liquide Global Engineering & Construction Solutions France
EPC CTR : EPC Contractor i.e. CTEP France SNC
CPY : Company i.e. QatarGas
RQF : Responsable Qualité fabrication
TBD : N° de document à venir / *Document reference to be updated*
ON/TPIA : Organisme notifié/ third party inspection agency
H : Hold point: Point d'arrêt, Avis préalable Exigé, Présence Obligatoire /
Notification required, presence mandatory
W : Witness Point: Point d'observation, Notification obligatoire, la Présence n'est pas Obligatoire /
Notification Required, presence not mandatory
SW : Spot Witness - Aperçu des activités sélectionnées auxquelles les inspecteurs AL peuvent
participer (selon eux) /
Overview of selected activities where AL inspectors may participate (of his own)
R : Revue Documentaire / *Document Review*
NA : Non applicable / *Not Applicable*

(*) : According to ALAT confidentiality policy, the mentioned documents are available only at ALAT site. Inspection report will be without dimension.

() : Materials are not communicated**

(*) : Only French version is available**

Note

Le code de construction ASME (American Society of Mechanical Engineers) Section VIII division 1 a été écrit pour la conception, la fabrication et la certification d'appareils à pression.

Ce code de construction est le plus ancien et le plus mature, il est reconnu dans l'industrie de part le monde comme une référence pour produire des équipements de qualité supérieure.

Pour cette raison, Air Liquide a choisi d'utiliser les règles applicables de ce code pour les turbines, même si ce type d'équipement n'entre pas dans le scope du code ASME.

The American Society of Mechanical Engineers (ASME) Section VIII division 1 construction code was written for the design, fabrication and stamping of pressure vessels.

This construction code is the older and the most mature in the pressure vessels domain, and worldwide industry has recognized it as a reference to provide top quality equipment.

For this reason, Air Liquide has chosen to use the applicable rules of this standard for the turbo expanders, even if this type of equipment is not in the scope of ASME code.

Enregistrement des noms, initiales, signatures, tampons des signataires de ce PIT

Registration of names, initials, signatures, stamps of this ITP signatories

[illegible]

 Business Confidential		Plan d'inspections et de Tests ITP - Inspection and Test Plan										R2-05-F38 (3)		
		HeRU 4 : Helium Liquefier and Distribution System C4168-IP-104(2)												
TC6-500 T4		Projet : PRJ000614 Project:		N ° de Fabrication :	Equipped part : ALAT 2301520 (criticality level IIc)							Code Racine : C4168 Root Ref. :		
	Description Opération	APPLICABLE SPEC./PROCEDURE/CODE (including release number)	confid entia lity	Fabrication et contrôle Manufacturing and Inspection Results								Enregistrement Record		Observations
				Coordinateur production Production coordinator		AQ	AL GECS	EPC CTR	CPY	Acceptance criteria				
				Initials	Visa	Visa	Visa	Visa	Visa					
0 Préalables généraux Prerequisite Actions														
0.1.1	Certificats matière Material certificates	ASME II RG6-S-00-1520-009	(*) (**)		H	R	R	R	R	NA		no material certificates provided for cartridges		
0.1.2	Qualification des soudeurs Welder qualification	ASME IX			H	R	R	R	R	NA				
0.1.3	Cahier de soudage (QMOS/DMOS) Welding book (WPS/PQR)	C4168-WB-114 C4168-WB-124 C4168-WB-134	(*)		H	R	R	R	R	NA				
0.1.4	Code construction Building code	ASME VIII			H	R	R	R	R	NA				
0.1.5	Qualification opérateur CND NDT coperator qualification	ISO 9712 level II			H	R	R	R	R	NA				
0.1.6	Réunion d'inspection avant lancement PIM: Pre-Inspection Meeting				H	H	H	H	H	NA	2023-CHPY-PIM-CRYO-01			
Données projet Project data														
0.1.7	Pression de design du projet Design Pressure of the project	30 b(g)			R	R	R	R	R	NA				
0.1.8	Pression du test d'épreuve pneumatique (fonction de la norme) Pneumatic test pressure (according to standard)	33 b(g)			R	R	R	R	R	NA				

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 Business Confidential		Plan d'inspections et de Tests ITP - Inspection and Test Plan										R2-05-F38 (3)	
		HeRU 4 : Helium Liquefier and Distribution System										C4168-IP-104(2)	
TC6-500 T4		Projet : PRJ000614 Project:		N ° de Fabrication :	Equipped part : ALAT 2301520 (criticality level IIc)							Code Racine : C4168 Root Ref. :	
	Description Opération	APPLICABLE SPEC./PROCEDURE/CODE (including release number)	confidentiality	Fabrication et contrôle Manufacturing and Inspection Results									
				Coordinateur production Production coordinator		AQ	AL GECS	EPC CTR	CPY	Acceptance criteria	Enregistrement Record	Observations	
				Initials	Visa	Visa	Visa	Visa	Visa				
10	Corps mécanique Mechanical assembly												
10.1 Levée des préalables Prerequisite Actions													
	Note: En référence à l'ASME VIII div 1, U-1(c)(2)(c), le corps mécanique est exclu du champ d'application de l'ASME. Note: Acc. To ASME VIII div 1, U-1(c)(2)(c) mechanical body is out of ASME scope.												
10.1.1	Vérifier que tous les documents demandés sont bien disponibles Check that each required file is available	/	(*)		H	R	R	R	R	NA		H: during final inspection	
10.2 Fabrication Fabrication													
10.2.1	Mesure de PMI - items & soudures (si techniquement possible) PMI Measurement - items & welds (when technically possible)	STD_G&C_350 RG6-S-00-1520-007	(*)		H	SW	SW	SW	SW	NA	PV		
10.2.2	Soudage Welding	C4168-WB-114	(*)		H	SW	SW	R	R	NA	C4168-WB-114	OF	
10.2.3	Contrôle visuel -100%- Visual inspection -100%-	R2-05-P13	(*)		H	R	R	R	R	ASME V	PV	(VT)	
10.2.4	Radiographie des soudures W101 et W102 Welding radiographic examination W101 et W102	R2-05-P19	(*)		H	R	R	R	R	ASME V	PV	(Rx) no PT exam as per deviation n°17	
10.3 Tests Tests													
10.3.1	Test Hydraulique Hydraulic Test	C4444-PO-066 1.5 x Design Pressure = 75 b	(*)		H	W	R	R	R	no crack, split or leak	PV	internal test prior machining/welding	
10.3.2	Test d'étanchéité Leak test	C4444-PO-069	(*)		H	W	R	R	R	Qf < 1.10-9 Pa.m3/s (1.10-8 mBar.l/sec) snifing : Qf < 1.10-6 Pa.m3/s (10-5 mBar.l/sec)	PV	internal test prior machining/welding (LT)	
10.3.3	Contrôle dimensionnel Dimensionnall checking	C4168E014A	(*)		H	SW	R	R	R	As per drawing	ALAT know how : no control report supplied	internal test prior machining/welding only for customer interfaces	

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20	Corps chaudronné <i>Welding assembly</i>											
20.1	Levée des préalables <i>Prerequisite Actions</i>											
20.1.0	Vérifier que tous les documents demandés sont bien disponibles <i>Check that each required file is available</i>				H	R	R	R	R			H: during final inspection
20.2	Fabrication <i>fabrication</i>											
20.2.1	Mesure de PMI - Items & soudures (si techniquement possible) <i>PMI Measurement - items & welds (when technically possible)</i>	STD_G&C_350 RG8-S-00-1520-007	(*)		H	SW	SW	SW	SW	NA	PV	
20.2.2	Soudage <i>Welding</i>	C4168-WB-124	(*)		H	SW	SW	R	R	NA	C4168-WB-124	OF
20.2.3	Contrôle visuel -100%- <i>Visual inspection -100%-</i>	R2-05-P13	(*)		H	R	R	R	R	ASME V	PV	(VT)
20.2.4	Radiographie 100% Préfabrication et intégration des lignes HP/BP <i>Radiographic examination 100% of HP & BP lines</i>	R2-05-P19	(*)		H	R	R	R	R	ASME V	PV	(Rx) no PT exam as per deviation n°17
20.2.5	Décapage passivation <i>Pickling and passivation</i>	D4444-SP-020	(*)		H	R	R	R	R	NA		
20.3	Tests <i>Tests</i>											
20.3.1	Test Hydraulique <i>Hydraulic Test</i>	C4444-PO-072 1,5 x Design Pressure = 75 bars (g)	(*)		H	W	W	W	W	no crack, split or leak	PV	
20.3.2	Test d'étanchéité <i>Leak test</i>	C4444-PO-075	(*)		H	W	W	W	W	Qf < 1,10-9 Pa.m3/s (1.10-8 mBar.l/sec) Sniffing : Qf < 1,10-6 Pa.m3/s (10-5 mBar.l/sec)	PV	(LT)
20.3.3	Contrôle dimensionnel <i>Dimensionnal checking</i>	C4168E042A	(*)		H	SW	SW	SW	SW	as per drawing +/-2 mm axial distance HP/BP	ALAT know how : no control report supplied	Only for customer interfaces

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Plan d'inspections et de Tests ITP - Inspection and Test Plan

R2-05-F38 (3)

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HeRU 4 : Helium Liquefier and Distribution System

C4168-IP-104(2)

TC6-500 T4

Projet : PRJ000614
Project:

N ° de
Fabrication :

Equipped part : ALAT 2301520 (criticality level IIc)

Code Racine : C4168
Root Ref. :

	Operation Description	APPLICABLE SPEC./PROCEDURE/CODE (including release number)	confid entiali ty	Fabrication et contrôle Manufacturing and Inspection Results						Acceptance criteria	Enregistrement Record	Observations
				Coordinateur production Production coordinator		AQ	AL GECS	EPC CTR	CPY			
				Initials	Visa	Visa	Visa	Visa	Visa			
30	Corps équipé <i>Equiped assembly</i>											
30.1	Levée des préalables <i>Prerequisite Actions</i>											
30.1.1	Vérifier que tous les documents demandés sont bien disponibles <i>Check that each required file is available</i>				H	R	R	R	R	NA		H: during final inspection
30.1.2	Instrumentation (Vannes, PT, tubes, panneau de controle) <i>Instrumentation (valves, PT, tubing, control panel, ...)</i>	68-14-PR-PID-01100			H	R	R	R	R			
30.2	Fabrication <i>fabrication</i>											
30.2.1	Mesure de PMI - items & soudures (si techniquement possible) <i>PMI Measurement - items & welds (when technically possible)</i>	STD_G&C_350 RG6-S-00-1520-007	(*)		H	SW	SW	SW	SW	NA	PV	
30.2.2	Soudage <i>Welding</i>	C4168-WB-134	(*)		H	SW	SW	R	R	NA	C4168-WB-134	OF
30.2.3	Contrôle visuel -100%- <i>Visual inspection -100%-</i>	R2-05-P13	(*)		H	R	R	R	R	ASME V	PV	(VT)
30.2.4	Radiographie 5% Préfabrication et 10% intégration des lignes selon ASME B31-3 <i>Radiographic examination 5% of warm line and 10% of integration according to ASME B31-3</i>	R2-05-P19 ING119-M	(*)		H	R	R	R	R	ASME V	PV	% of all the warm line and by welder (Rx) Tubing
30.2.5	Propreté, visite interne (avec photo internes : HP, BP, bride) <i>Cleanliness, internal visit (including picture internal: HP, BP, flange)</i>	FAB-001	(*)		H	R	R	R	R	as per procedure	PV	
30.3	Tests <i>tests</i>											
30.3.1	Conformité PID <i>P&ID conformtity</i>	68-14-PR-PID-01100 R2-05-MOP11	(*)		H	R	R	R	R	NA	PV	
30.3.2	Tests finaux (épreuve pneumatique) <i>Final tests (pneumatic test)</i>	D4444-PO-008 C4444-PO-041	(*)		H	W	W	W	W	30 min under 33 b(g) Nitrogen	PV	53125-RE-050-313F501 final test procedure to be submitted in document class 3 for information
30.3.3	Tests finaux (tests étanchéité Helium) <i>Final tests (Helium leak test)</i>		(*)		H	W	W	W	W	10 min under 27 b(g) Helium acceptance as per §6.1 of C4444-PO-041	PV	53125-RE-050-313F501 final test procedure to be submitted in document class 3 for information
30.3.4	Tests finaux (tests fonctionnels) <i>Final tests (functional tests)</i>		(*)		H	W	W	W	W	NA	PV	53125-RE-050-313F501 final test procedure to be submitted in document class 3 for information (LT sur CDS)
30.3.5	Contrôle dimensionnel <i>Dimensionnall checking</i>	C4168E041A	(*)		H	R	R	R	R	as per drawing	PV	ALAT internal check
30.3.6	Peinture <i>Painting</i>	RG6-S-00-13A0-004	(*)		H	R	R	R	R	according to table 3 of specification	PV	
30.3.7	Inspection finale (visuel, dimensionnel, propreté inclus photo bride, peinture, tagging, PID) Revue documentaire QC <i>Final inspection (visual, dimensional, clenliness including flange face picture, painting, tagging, PID) QC Documentation review</i>				H	H	H	H	H		PV	

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 Business Confidential		Plan d'inspections et de Tests ITP - Inspection and Test Plan										R2-05-F38 (3)	
		HeRU 4 : Helium Liquefier and Distribution System										C4168-IP-104(2)	
TC6-500 T4		Projet : PRJ000614		N ° de Fabrication :	Main cartridge : TC6-639-HI1 (criticality level IIc)						Code Racine : C4168 Root Ref. :		
	Operation Description	APPLICABLE SPEC./PROCEDURE/CODE (including release number)	confidentiality	Fabrication et contrôle Manufacturing and Inspection Results									
				Coordinateur production Production coordinateur		AQ	AL GECS	EPC CTR	CPY	Acceptance criteria	Enregistrement Record	Observations	
				Initials	Visa	Visa	Visa	Visa	Visa				
40	Cartouche Cartridge												
40.1 Tests Tests													
40.1.1	Equilibrage Balancing		(**)		R	R	NA	NA	NA		ALAT know how : no control report supplied		
40.1.2	Tests mécanique et cryogénique avec survitesse Mechanical and cryogenic tests with overspeed	F2610-PO-007			H	W	W	W	W	as per procedure	PV	53125-RE-050-313F753	
50	Ensemble Turbine Turbine assembly												
50.1 Spécificité client avant envoi Customer specificity before release													
50.1.1	Preservation Preservation	RG6-S-00-13A0-007 NFEF-60-00-SP-TS-NA-009(checklist only)			H	W	W	W	W	as per specification	Checklist		
50.1.2	Certificat d'inspection avant envoi ICN : Inspection Completion Note / IRN : Inspection Release Note	RG6-S-00-1520-001			H	H	H	H	R	NA	ICN/IRN		
50.1.3	Préparation pour expédition et emballage, inspection Preparation for shipment & packing inspection	AK2-S-00-1400-021			H	H	W	W	W	as per specification	Checklist		
50.1.4	Revue du MRB MRB review	NFEF-60-00-SP-TS-NA-001			H	R	R	R	R	NA	MRB	MRB submitted for approval	

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 Business Confidential			Plan d'inspections et de Tests ITP - Inspection and Test Plan										R2-05-F38 (3)	
			HeRU 4 : Helium Liquefier and Distribution System C4168-IP-104(2)											
TC6-500 T4			Projet : PRJ000614 Project:		N ° de Fabrication :	Spare cartridge : TC6-642-HI1 (criticality level IIc)						Code Racine : C4168 Root Ref. :		
	Operation Description		APPLICABLE SPEC./PROCEDURE/CODE (including release number)	confidentiality	Fabrication et contrôle Manufacturing and Inspection Results									
Coordinateur production Production coordinateur					AQ	AL GECS	EPC CTR	CPY	Acceptance criteria	Enregistrement Record	Observations			
Initials					Visa	Visa	Visa	Visa				Visa		
60	Cartouche Cartridge													
60.1	Tests Tests													
60.1.1	Equilibrage Balancing			(**)		R	R	NA	NA	NA		ALAT know how : no control report supplied		
60.1.2	Tests mécanique et cryogénique avec sur vitesse Mechanical and cryogenic tests with overspeed		F2610-PO-007			H	W	W	W	W	as per procedure	PV	53125-RE-050-313F753	

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